

1. Atomic Structure & Electron Configuration

Page 1 of 2 • Official WAEC & JAMB Syllabus Guide

Subatomic Particles

- Protons: Mass = 1 amu, Charge = +1, located in nucleus.
- Neutrons: Mass = 1 amu, Charge = 0, located in nucleus.
- Electrons: Mass $\frac{1}{1840}$ amu, Charge = -1, orbit in shells (K, L, M, N).

Valence & Octet Rule

Atoms tend to gain, lose, or share electrons to achieve a stable electronic configuration of 8 valence electrons (or 2 for Helium).

2. Types of Chemical Bonds

Page 2 of 2 • Official WAEC & JAMB Syllabus Guide

Electrovalent (Ionic) Bonding

Occurs between metals and non-metals via complete transfer of electrons (e.g. NaCl, MgO).
Properties: High melting/boiling points, conduct electricity in molten or aqueous state.

Covalent & Coordinate Bonding

Formed when atoms share electron pairs (e.g. H_2O , CH_4 , NH_3).
Dative (Coordinate) bond: Both shared electrons are provided by one atom.